

COMPUTER ORGANIZATION AND ARCHITECTURE								
II B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS08	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100
COURSE OUTCOMES:								
At the end of the course students will be able to:								
1. Identify various components of computer and their interconnection								
2. Identify basic components and design of the CPU: the ALU and control unit.								
3. Compare and select various Memory devices as per requirement.								
4. Compare various types of IO mapping techniques								
5. Critique the performance issues of cache memory and virtual memory								
UNIT-I								
STRUCTURE OF COMPUTERS: STRUCTURE OF COMPUTERS: Computer types, Functional units, Basic operational concepts, Von Neumann Architecture, Bus Structures, Software, Performance, Multiprocessors and Multicomputer, Data representation, Fixed and Floating point. COMPUTER ARITHMETIC: Addition and Subtraction, Multiplication and Division algorithms, Floating-point Arithmetic Operations, Decimal arithmetic operations								
UNIT-II								
BASIC COMPUTER ORGANIZATION AND DESIGN: BASIC COMPUTER ORGANIZATION AND DESIGN: Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Timing and Control, Memory-Reference Instructions, Input-Output and interrupt. Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC) Reduced Instruction Set Computer (RISC), CISC vs RISC								
UNIT-III								
REGISTER TRANSFER AND MICRO-OPERATIONS: REGISTER TRANSFER AND MICRO-OPERATIONS: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit. MICRO-PROGRAMMED CONTROL: Control Memory, Address Sequencing, Micro-Program, Design of Control Unit								
UNIT-IV								
MEMORY SYSTEM: MEMORY SYSTEM: Memory Hierarchy, Semiconductor Memories, RAM(Random Access Memory), Read Only Memory (ROM), Types of ROM, Cache Memory, Virtual memory, Secondary Storage, RAID.								
UNIT-V								
MEMORY SYSTEM: INPUT OUTPUT: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA. MULTIPROCESSORS: Characteristics of multiprocessors, Interconnection structures, Inter Processor Arbitration, Inter processor Communication and Synchronization, Cache Coherence								
TEXT BOOKS:								
1. M. Morris Mano (2006), Computer System Architecture, 3rd edition, Pearson/PHI, India.								
2. John P. Hayes (1998), Computer Architecture and Organization, 3rd edition, Tata McGrawHill								
REFERENCE BOOKS:								
1. Carl Hamacher, Zvonks Vranesic, SafeaZaky (2002), Computer Organization, 5th edition,McGraw Hill, New Delhi, India.								
2. William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersey.								
3. Anrew S. Tanenbaum (2006), Structured Computer Organization, 5th edition, Pearson Education Inc.								